NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

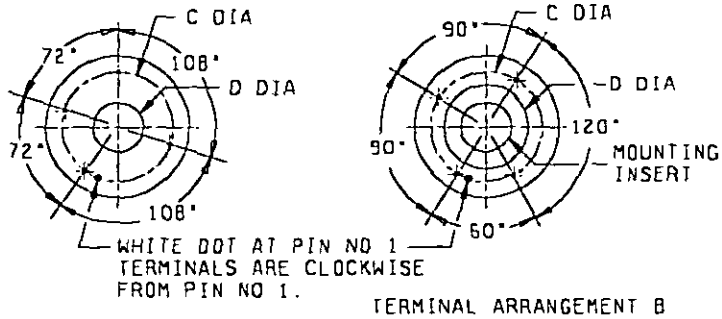
MIL-PRF-27/356C
8 April 1992
SUPERSEDING
MIL-T-27/356B
25 January 1985

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS AND INDUCTORS
(AUDIO, POWER AND HIGH-POWER PULSE),
INDUCTORS, POWER, TF5S04ZZ

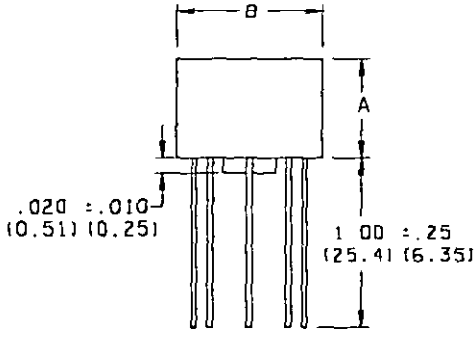
This specification is approved for use by all Departments and Agencies of the Department of Defense.

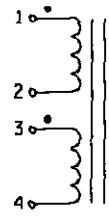
The requirements for acquiring the transformer described herein shall consist of this specification and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-T-27.



TERMINAL ARRANGEMENT A

TERMINAL ARRANGEMENT B





CIRCUIT DIAGRAM

FIGURE 1. Dimensions and configuration.

© denotes changes

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Dash number	A Max	B Max Dia	C ±.010	Insert	Terminal arrangement	Terminal diameter inches ±.002 (C)
01,06,11,16, 21,26,31,36, 41,46,51,56, 61	280 (7 11)	410 (10 41)	260 (6 6)		A	012
02,07,12,17, 22,27,32,37, 42,47,52,57, 62	.340 (8 63)	.500 (12 7)	.350 (8 89)		A	.012
03,08,13,18, 23,28,33,38, 43,48,53,58, 63	415 (10 54)	630 (16 0)	440 (11 17)		A	.016
04,09,14,19, 24,29,34,39, 44,49,54,59	500 (12 70)	800 (20.32)	560 (14 22)		B	016
05,10,15,20, 25,30,35,40, 45,50,55,60	635 (16.12)	950 (24 13)	670 (17 0)	138-32 UNC-2B .200 Min deep	B	020

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for general information only
(C) 3 Marking shall be on the side or on top
4 For series connection join terminals 2 and 3, for parallel connection join terminals 1 and 3, 2 and 4
5 Metric equivalents are in parentheses

FIGURE 1 Dimensions and configurations - Continued

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TABLE I Electrical ratings

Dash number	1/ Series inductance (mH)	Series dc current (max) (amps)	Series dc resistance ±25% (ohms)	1/ Parallel inductance (mH)	Parallel dc current (max) (amps)	Parallel dc resistance ±25% (ohms)
01	20	.06	54	5	12	13 5
02	20	10	29	5	.20	7 25
03	20	14	20	5	.28	5.0
04	20	.22	12.4	5	44	3 1
05	20	.29	10.2	5	58	2 55
06	15	07	40.5	3.75	14	10 125
07	15	.12	22	3.75	.24	5.5
08	15	16	15	3 75	.32	3 75
09	15	.25	9.3	3.75	.50	2.325
10	15	33	7 7	3 75	66	1 925
11	10	08	27	2.5	.16	6 75
12	10	14	14 6	2 5	.28	3 65
13	10	20	10	2 5	40	2 5
14	10	31	6.2	2.5	62	1 55
15	10	.40	5 1	2.5	.80	1.275
16	7.5	10	20.3	1.88	20	5 075
17	7 5	17	11	1 88	34	2 75
18	7.5	.24	7.5	1.88	48	1 875
19	7 5	.38	4 7	1.88	76	1.175
20	7.5	.49	3 8	1.88	98	95

See footnote at end of table

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TABLE I Electrical ratings - Continued

Dash number	1/ Series inductance (mH)	Series dc current (max) (amps)	Series dc resistance ±25% (ohms)	1/ Parallel inductance (mH)	Parallel dc current (max) (amps)	Parallel dc resistance ±25% (ohms)
21	5.0	.12	13 5	1.25	.24	3 375
22	5 0	.21	7 3	1.25	.42	1.825
23	5.0	.30	5	1.25	60	1 25
24	5.0	.47	3 1	1.25	.94	.775
25	5 0	6	2 6	1 25	1 2	65
26	5 0	16	8 1	.75	.32	2 025
27	3	27	4 4	.75	.54	1 1
28	3	38	3	.75	.76	75
29	3	6	1 9	75	1.2	475
30	3	.78	1 5	.75	1.56	.375
31	2	19	5 4	5	38	1 35
32	2	33	2.9	5	66	725
33	2	47	2	5	.94	5
34	2	74	1 3	5	1 48	325
35	2	95	1	5	1 9	25
36	1	.27	2 7	.25	54	675
37	1	47	1 5	.25	94	375
38	1	.66	1	25	1 32	25
39	1	1	62	25	2	155
40	1	1.3	5	25	2 6	125

See footnote at end of table

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TABLE I Electrical ratings - Continued

Dash number	1/ Series inductance (mH)	Series dc current (max) (amps)	Series dc resistance ±25% (ohms)	1/ Parallel inductance (mH)	Parallel dc current (max) (amps)	Parallel dc resistance ±25% (ohms)
41	750	31	2	1875	.62	5
42	.750	.54	1	.1875	1.08	.25
43	750	76	75	.1875	1.52	1875
44	.750	1.2	.47	.1875	2.4	.1175
45	.750	1 6	38	1875	3.2	095
46	500	38	1 3	125	.76	.325
47	500	66	66	125	1 32	165
48	.500	93	5	125	1 86	125
49	500	1 47	31	125	2 94	.0775
50	500	1 9	25	125	3 8	0625
51	250	.53	66	0625	1.06	165
52	250	93	33	0625	1 86	0825
53	250	1 3	.25	0625	2 6	0625
54	250	2 08	16	.0625	4.16	04
55	.250	2 7	13	0625	5.4	.0325
56	100	78	.30	.025	1 56	075
57	100	1 35	17	025	2 7	043
58	.100	1 95	.11	.025	3.9	.028
59	.100	3 08	07	.025	6.16	.018
60	.100	3 88	.06	.025	7.76	.015

See footnote at end of table

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TABLE I. Electrical ratings - Continued

Dash number	1/ Series inductance (mH)	Series dc current (max) (amps)	Series dc resistance ±25% (ohms)	1/ Parallel inductance (mH)	Parallel dc current (max) (amps)	Parallel dc resistance ±25% (ohms)
61	.040	1.18	.14	.01	2 36	035
62	.040	2.03	.07	.01	4.06	.018
63	040	2 68	.06	.01	5.36	015

1/ The inductance must be at least 80 percent of the listed value when measured with rated dc current at 0 1 V rms, 10 kHz

REQUIREMENTS (When numbers in parentheses, i e , (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings See table I

Design and construction

Dimensions and configurations See figure 1

Duty cycle Continuous

Case Epoxy

Terminals Printed circuit, solderable

Weight 1.25 ounces, maximum.

Altitude 75,000 feet, maximum.

Operating temperature range -55°C to +130°C

Terminal strength MIL-STD-202, method 211, test condition A, 2 pounds

Dielectric withstanding voltage

At sea level 200 volts rms.

At reduced barometric pressure 100 volts rms

Vibration (high frequency) MIL-STD-202, method 204

Magnetic shield All units are magnetically shielded

Temperature rise 30°C with 1 volt rms, rated dc current as specified in table I at an ambient temperature of 100°C, maximum

Marking location See figure 1

Part or Identifying Number (PIN) M27/356-(dash number from table I)

Qualification Qualification testing and approval to M27/356-01 shall be sufficient to grant qualification approval to M27/356-01 through M27/356-63

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CONCLUDING MATERIAL

Custodians
  Army - ER
  Navy - EC
  Air Force - 85

Review activities
  Army - MI
  Navy - OS
  Air Force - 17, 99
  DLA - ES

User activities
  Army - AR
  Navy - MC, AS
  Air Force - 19

Preparing activity
  Army - ER

Agent
  DLA - ES

(Project 5950-0795)

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